Types:

→ Wired
- → UTP
- → Fiber

→ Wireless

AUTO MDIX:

→ It can connect any cables into straight or cross even it is a same functional device. Eg: PC & Router or different functional device.

PC		PC
+T	———	+T → +R
−T	———	−T → −R
+R	———	+R → +T
−R	———	−R → −T

→ This is done by AUTO MDIX.

Day - 3 IP & MAC

IP → Internet Protocol - Location of a device

MAC → Media access control - Identification of a device

MAC:

↳ Physical address - Because it comes with the NIC card during manufacturing

↳ It is in hexadecimal format which has 4 bits & contains 12 hexadecimal.

So $\boxed{12 \times 4 = 48 \text{ bits}}$

Eg: 1C-C1-DE-33-8D-14 → 00011100 - 11000001 - 11011110 - 00110011 - 10001101 - 00010100

MAC has 48 bits

24 bits → Vendor ID

24 bits → User ID

J. Bagavathi

↙ vendor ↘ user

IP:

↳ It is a logical address → configured by us

↳ It has dotted decimal format

↳ Contains 4 octet (1 octet = 8 bits)

⇒ $\boxed{8 \times 4 = 32 \text{ bits}}$

IP = Network + Host

IPV4 structure :-
↳ Version 4

1st

128	64	32	16	8	4	2	1

2nd

128	64	32	16	8	4	2	1

3rd

128	64	32	16	8	4	2	1

4th

128	64	32	16	8	4	2	1

0.0.0.0 : Starting IP

↕

255.255.255.255 : Ending IP

2-bit Binary :

2	1
0	0
0	1
1	0
1	1

3-bit Binary :

4	2	1
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

5-bit Binary:

16	8	4	2	1	
0	0	0	0	0	→ 0
0	0	0	0	1	
0	0	0	1	0	→ 2
0	0	0	1	1	
0	0	1	0	0	→ 4
0	0	1	0	1	
0	0	1	1	0	
0	0	1	1	1	→ 8
0	1	0	0	0	
0	1	0	0	1	
0	1	0	1	0	
0	1	0	1	1	
0	1	1	0	0	
0	1	1	0	1	
0	1	1	1	0	
0	1	1	1	1	
1	0	0	0	0	→ 16
1	0	0	0	1	
1	0	0	1	0	
1	0	0	1	1	
1	0	1	0	0	
1	0	1	0	1	
1	0	1	1	0	
1	0	1	1	1	→ 23

Different classes of IP.

CLASS	RANGE	PORTION
A	0-127	N.H.H.H
B	128-191	N.N.H.H
C	192-223	N.N.N.H
D	224-239	Multicast
E	240-255	Broadcast R & D

} Invalid I.P

Eg:

1) 72.0.0.86 - A - N.H.H.H
2) 172.10.234.254 - B - N.N.H.H
3) 192.10.192.51 - C - N.N.N.H

Network - Street No.
Host - Door No.

$\underset{N/W}{50}.\underbrace{0.0.10}_{Host}$

Types of communication:

Unicast : One to One
Multicast : One to group
Broadcast : One to all

8 8 8 8
192.192.192.192
N . N . N . H

No. of network bits : 24
No. of host bits : 8
No. of network octet : 3
No. of host octet : 1